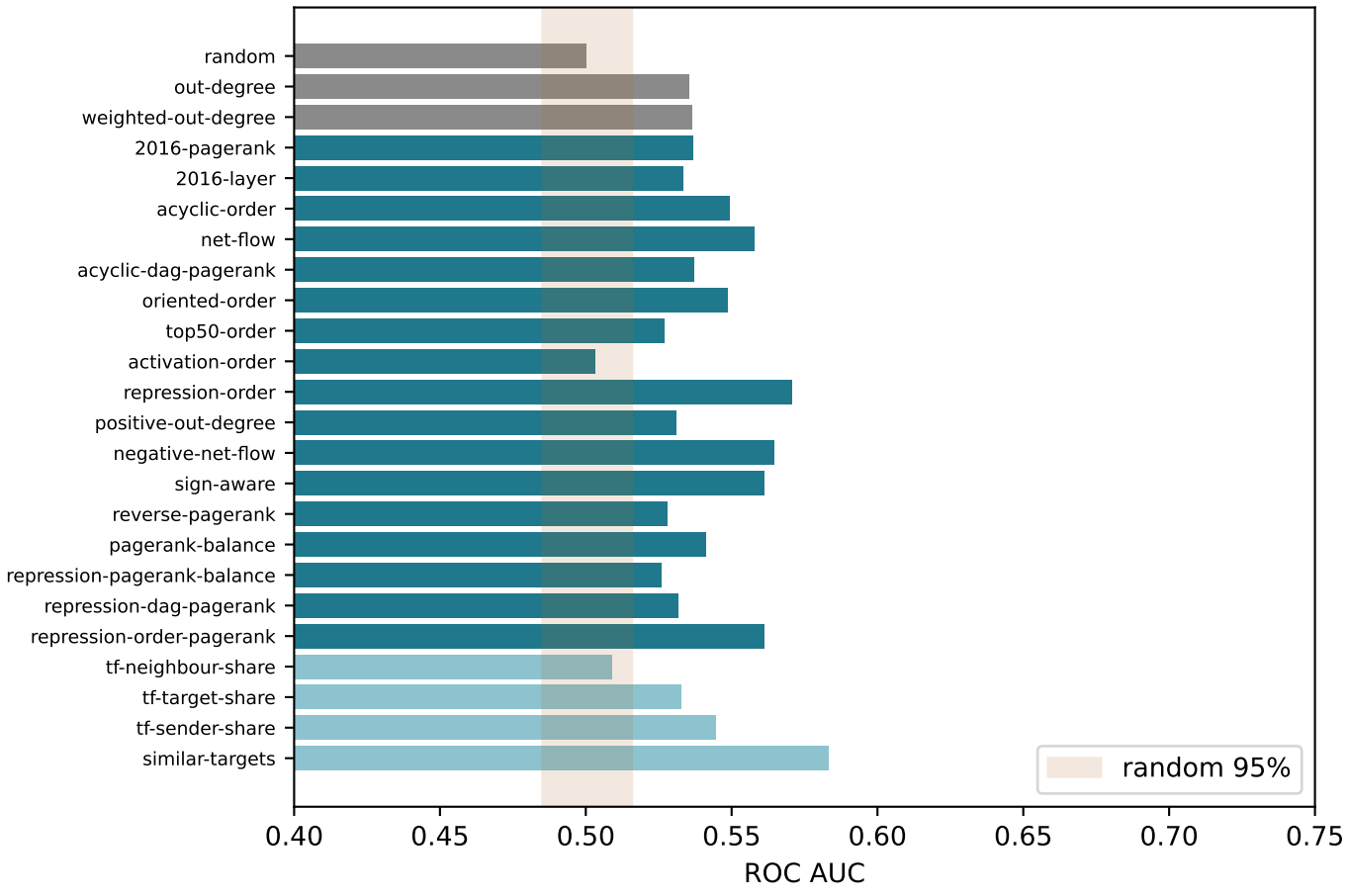
Regulator ranking against reference 'tfs'

2026-11-08 · 10,155 genes ranked, 1,639 in tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	16.0 (10–23)	0.500 (0.485–0.516)				
out-degree	18	0.535	0.338	< 0.001	-0.001 (-0.002, +0.000)	0.170
weighted-out-degree	18	0.536	0.338	< 0.001	+0.000 (+0.000, +0.000)	
2016-pagerank	16	0.537	0.544	< 0.001	+0.001 (-0.021, +0.022)	0.964
2016-layer	15	0.533	0.649	< 0.001	-0.003 (-0.020, +0.013)	0.676
acyclic-order	21	0.549	0.114	< 0.001	+0.013 (+0.004, +0.022)	0.004
net-flow	19	0.558	0.248	< 0.001	+0.022 (+0.012, +0.031)	< 0.001
acyclic-dag-pagerank	23	0.537	0.038	< 0.001	+0.001 (-0.006, +0.008)	0.814
oriented-order	20	0.549	0.173	< 0.001	+0.013 (+0.003, +0.021)	0.006
top50-order	14	0.527	0.746	0.001	-0.009 (-0.030, +0.009)	0.320
activation-order	16	0.503	0.544	0.350	-0.033 (-0.050, -0.019)	< 0.001
repression-order	17	0.571	0.438	< 0.001	+0.035 (+0.021, +0.048)	< 0.001
positive-out-degree †	16	0.531	0.544	< 0.001	-0.005 (-0.012, +0.001)	0.094
negative-net-flow †	18	0.565	0.338	< 0.001	+0.028 (+0.017, +0.039)	< 0.001
sign-aware †	18	0.561	0.338	< 0.001	+0.025 (+0.016, +0.034)	< 0.001
reverse-pagerank †	18	0.528	0.338	< 0.001	-0.008 (-0.015, -0.002)	0.018
pagerank-balance †	19	0.541	0.248	< 0.001	+0.005 (-0.002, +0.013)	0.192
repression-pagerank-balance †	13	0.526	0.824	0.001	-0.010 (-0.020, -0.001)	0.032
repression-dag-pagerank †	16	0.532	0.544	< 0.001	-0.005 (-0.016, +0.006)	0.420
repression-order-pagerank †	29	0.561	< 0.001	< 0.001	+0.025 (+0.016, +0.035)	< 0.001
tf-neighbour-share ‡	29	0.509	< 0.001	0.132	-0.027 (-0.050, -0.007)	0.016
tf-target-share ‡	34	0.533	< 0.001	< 0.001	-0.004 (-0.023, +0.016)	0.666
tf-sender-share ‡	38	0.545	< 0.001	< 0.001	+0.008 (-0.012, +0.029)	0.436
similar-targets ‡	34	0.583	< 0.001	< 0.001	+0.047 (+0.027, +0.066)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).

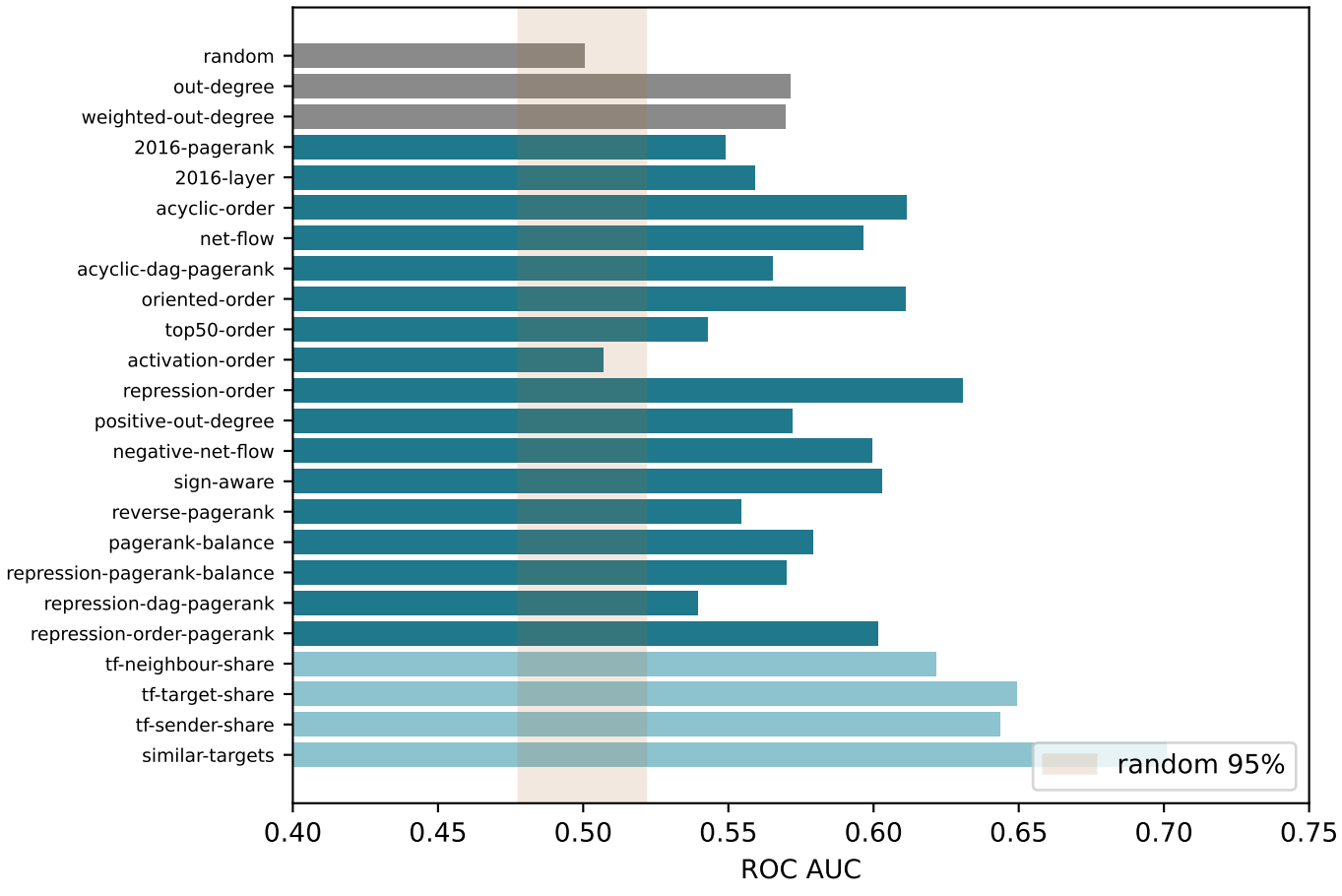


Regulator ranking against reference 'lambert-2018'

2026-11-08 · 10,155 genes ranked, 741 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	7.2 (3–13)	0.500 (0.478–0.522)				
out-degree	14	0.571	0.012	< 0.001	+0.000 (+0.000, +0.000)	
weighted-out-degree	14	0.570	0.012	< 0.001	-0.002 (-0.003, -0.000)	0.044
2016-pagerank	7	0.549	0.585	< 0.001	-0.022 (-0.052, +0.009)	0.180
2016-layer	7	0.559	0.585	< 0.001	-0.012 (-0.036, +0.011)	0.320
acyclic-order	12	0.611	0.048	< 0.001	+0.040 (+0.028, +0.052)	< 0.001
net-flow	12	0.596	0.048	< 0.001	+0.025 (+0.012, +0.038)	< 0.001
acyclic-dag-pagerank	13	0.565	0.027	< 0.001	-0.006 (-0.016, +0.004)	0.260
oriented-order	15	0.611	0.005	< 0.001	+0.039 (+0.027, +0.051)	< 0.001
top50-order	11	0.543	0.099	< 0.001	-0.029 (-0.056, -0.002)	0.030
activation-order	6	0.507	0.736	0.287	-0.065 (-0.088, -0.042)	< 0.001
repression-order	13	0.631	0.027	< 0.001	+0.059 (+0.042, +0.077)	< 0.001
positive-out-degree †	11	0.572	0.099	< 0.001	+0.001 (-0.007, +0.008)	0.896
negative-net-flow †	10	0.599	0.179	< 0.001	+0.028 (+0.014, +0.042)	< 0.001
sign-aware †	18	0.603	< 0.001	< 0.001	+0.032 (+0.020, +0.043)	< 0.001
reverse-pagerank †	10	0.554	0.179	< 0.001	-0.017 (-0.026, -0.008)	< 0.001
pagerank-balance †	13	0.579	0.027	< 0.001	+0.008 (-0.002, +0.018)	0.116
repression-pagerank-balance †	2	0.570	0.996	< 0.001	-0.001 (-0.013, +0.011)	0.840
repression-dag-pagerank †	5	0.540	0.855	< 0.001	-0.032 (-0.046, -0.016)	< 0.001
repression-order-pagerank †	20	0.602	< 0.001	< 0.001	+0.030 (+0.017, +0.044)	< 0.001
tf-neighbour-share ‡	48	0.622	< 0.001	< 0.001	+0.050 (+0.022, +0.077)	< 0.001
tf-target-share ‡	45	0.649	< 0.001	< 0.001	+0.078 (+0.052, +0.103)	< 0.001
tf-sender-share ‡	44	0.644	< 0.001	< 0.001	+0.072 (+0.046, +0.099)	< 0.001
similar-targets ‡	57	0.701	< 0.001	< 0.001	+0.129 (+0.100, +0.159)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,959	0.051 (1.50×)	2.56×	0.68×	0.68 (0.67)
resolve_reciprocals → drop_sinks → prune_triangles	31,959	0.009 (0.82×)	7.42×	0.79×	0.17 (0.16)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	26,831	0.009 (0.85×)	11.12×	0.84×	0.13 (0.13)
absolute	31,959	0.051 (1.50×)			
absolute → minimum_feedback_order	31,959	0.036 (1.22×)			
orient_reciprocals ratio=2.0 → absolute	31,959	0.031 (1.13×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	31,959	0.025 (1.01×)			
top_outgoing k=50 → absolute	31,959	0.008 (5.21×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,959	0.005 (3.62×)			
keep_sign sign=1 → absolute	31,874	0.038 (2.56×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,922	0.026 (2.26×)			
keep_sign sign=-1 → absolute	31,959	0.013 (0.68×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	31,959	0.013 (0.68×)			
keep_sign sign=-1	31,959	0.013 (0.68×)		0.68×	0.15 (0.15)

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	21,572	0.081 (2.34×)	4.48×	0.71×	0.81 (0.77)
resolve_reciprocals → drop_sinks → prune_triangles	21,572	0.012 (1.07×)	16.38×	0.99×	0.31 (0.26)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	19,024	0.011 (1.02×)	15.71×	1.01×	0.14 (0.14)
absolute	21,572	0.081 (2.34×)			
absolute → minimum_feedback_order	21,572	0.048 (1.61×)			
orient_reciprocals ratio=2.0 → absolute	21,572	0.041 (1.46×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	21,572	0.028 (1.09×)			
top_outgoing k=50 → absolute	21,572	0.010 (7.02×)			
top_outgoing k=50 → absolute → minimum_feedback_order	21,572	0.006 (4.42×)			
keep_sign sign=1 → absolute	21,566	0.067 (4.47×)			
keep_sign sign=1 → absolute → minimum_feedback_order	20,981	0.040 (3.40×)			
keep_sign sign=-1 → absolute	21,572	0.014 (0.71×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	21,572	0.013 (0.70×)			
keep_sign sign=-1	21,572	0.014 (0.71×)		0.71×	0.25 (0.25)

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	6,076	0.062 (1.91×)	3.46×	0.74×	0.67 (0.61)
resolve_reciprocals → drop_sinks → prune_triangles	6,076	0.009 (0.97×)	8.06×	0.92×	0.32 (0.30)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	5,311	0.009 (0.89×)	0.00×	0.89×	0.28 (0.28)
absolute	6,076	0.062 (1.91×)			
absolute → minimum_feedback_order	6,076	0.042 (1.53×)			
orient_reciprocals ratio=2.0 → absolute	6,076	0.035 (1.33×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	6,076	0.029 (1.20×)			
top_outgoing k=50 → absolute	6,076	0.011 (7.40×)			
top_outgoing k=50 → absolute → minimum_feedback_order	6,076	0.007 (5.54×)			
keep_sign sign=1 → absolute	6,069	0.048 (3.45×)			
keep_sign sign=1 → absolute → minimum_feedback_order	5,911	0.031 (2.93×)			
keep_sign sign=-1 → absolute	6,076	0.014 (0.74×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	6,076	0.013 (0.75×)			
keep_sign sign=-1	6,076	0.014 (0.74×)		0.74×	0.40 (0.40)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 10,100,294 edges (5,562,481 negative), largest strong component 10,137.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 10,100,294 edges (5,562,481 negative), largest strong component 10,137.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 2,493,517 edges (2,471,574 negative), largest strong component 9,984.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 2,493,517 edges (2,471,574 negative), largest strong component 9,984; DAG keeps 2,151,959 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 10,100,294 edges (0 negative), largest strong component 10,137; DAG keeps 8,319,970 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 10,100,294 edges (0 negative), largest strong component 10,137.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 10,100,294 edges (0 negative), largest strong component 10,137; DAG keeps 8,319,970 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 8,207,593 edges (0 negative), largest strong component 10,120; DAG keeps 7,395,637 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 500,081 edges (0 negative), largest strong component 8,664; DAG keeps 447,417 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 4,537,813 edges (0 negative), largest strong component 10,026; DAG keeps 3,605,450 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 5,562,481 edges (0 negative), largest strong component 9,758; DAG keeps 5,104,034 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 10,100,294 edges (5,562,481 negative), largest strong component 10,137.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 10,100,294 edges (5,562,481 negative), largest strong component 10,137.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 10,100,294 edges (5,562,481 negative), largest strong component 10,137.

reverse-pagerank: Highest reverse PageRank (raw graph)